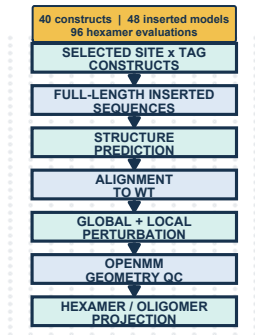
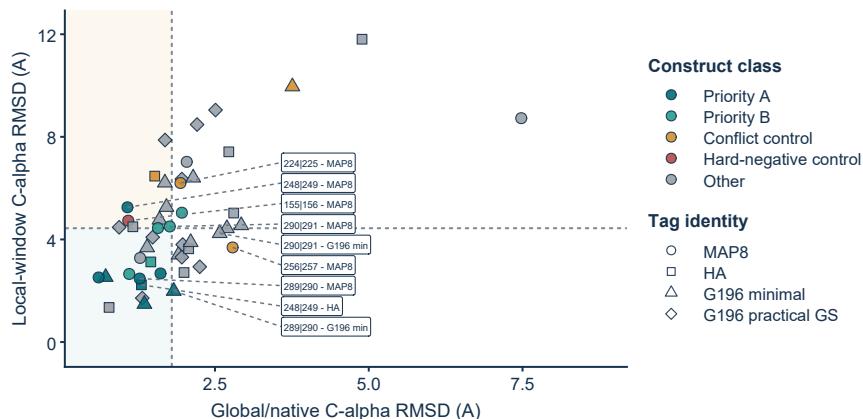


a**Explicit insertion-modeling workflow**

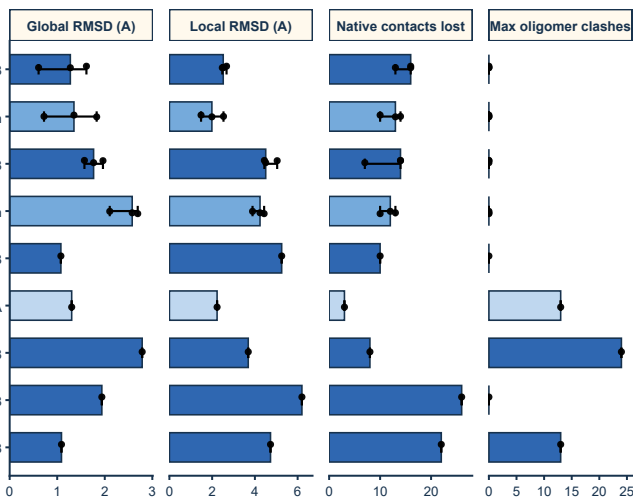
48/48 tagged models completed geometry QC

**b****Global structural perturbation landscape**

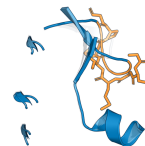
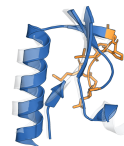
Each point is one inserted model; dashed medians are visual guides, not pass/fail cut-offs.

**c****Focal construct comparison**

Blue bars: medians; black dots: individual models; whiskers: min-max range.

Tag identity HA G196 minimal MAP8**d Representative WT overlays**

WT light grey; inserted chain blue; tag orange.

289|290 x MAP8 | lower local perturbation**290|291 x MAP8 | adjacent-junction contrast****248|249 x MAP8 | global/local discordance**